STAT 8480 Project Proposal

Our team consists of Jess Kotlarek, Nick Grieco, and Josh Mazurkiewicz, and our tentative project title is "Checking Out Before Checking In: Predicting Hotel Booking Cancellations." Our plan is to build a binary propensity model where we predict if a guest will cancel their hotel booking or not. Cancellations are not ideal for hotels as a cancelled reservation usually results in an empty room, and in our data, about 37% of hotel bookings are canceled. Our goal is to flag guests that are likely to cancel. A hotel could then send those guests reminders or require deposits for the riskiest ones. With these insights, hotels could also have a more realistic estimate of how much to overbook based on real data rather than a gut feeling. This is a yes/no predictions problem, so this is why we're building a propensity model.

The target is "is_canceled," which takes value 1 if the booking was canceled, or 0 if the guest showed up. We'll split the data into training, validation, and potentially a test set. We'll try logistic regression and a decision tree model, and maybe some ensemble models like a random forest or xgboost if time and interpretability permit. We will evaluate our results using ROC AUC or AIC/BIC. We'll also consider misclassification rate, precision, and recall, particularly from the validation data, focusing on the right metrics for the goal. Assuming a hotel would keep profit in mind, we would want to prioritize checking false negatives, so this might focus on having high recall so we can catch as many cancellations as possible.

We will use the Hotel Booking Demand dataset from Antonio, de Almeida and Nunes (2019) (https://www.sciencedirect.com/science/article/pii/S2352340918315191#f0010). It has 119,390 real bookings from two hotels in Portugal: a resort hotel (40,060 bookings) and a city hotel (79,330 bookings), for stays between July 2015 and August 2017. There are 32 columns: the target variable and 31 potential inputs, 25 of which are usable. The input variables consist of things like lead time, nightly rate, length of stay, number of guests (separated by baby/child/adult) and past cancellations with categories like hotel type, meal plan, country of visitor, market segment, booking channel, room type, deposit type, and customer type. The data is easily accessible on the TidyTuesday project in GitHub (https://github.com/rfordatascience/tidytuesday/blob/main/data/2020/2020-02-11/) and we can load it into R as one file.

The data will definitely need preparation before Phase 1: Data Audit and Descriptive Analysis. We will have to handle missing values in children, country, agent and company; check for weird values like extreme nightly rates or bookings with zero guests; group categories with too many levels such as country (one way we can do this is by building a base decision tree and see how the categories are grouped); and maybe adding a few new variables like total nights and total guests. Related research is limited but the dataset is described in the Antonio, de Almeida and Nunes (2019) article. The same authors also predicted cancellations using different data from four resort hotels in "Predicting hotel booking cancellations to decrease uncertainty and increase revenue (https://www.tmstudies.net/index.php/ectms/article/view/1000/pdf_51). They hit over 90% accuracy, which gives us something to compare against.

In class notes:

- Data must have >= 1,000 observations
- Data must have >= 10 potential (viable) input variables

Data

Requirements:

- Your project proposal should be approximately 1 page long, single spaced. The purpose of the proposal is to make sure that you are on the right track and to give me enough information so that I can give you useful feedback.
- In your proposal you should cover the following items:
    - Preliminary title and list of students working on the project
    - Brief description of what you plan to do.
        - What problem are you trying to solve?
        - How do you formulate the problem as a data mining problem? What exactly are you trying to predict and how will you evaluate your results? How will you know if your results are good? What can you compare them to? It is critical that your problem is well-defined.
        - What data sets do you plan to use? If you must do significant work to get the data or convert it into the proper format, then describe the process and approximate effort required. How many observations are in the data set? How many variables?
        - If sufficiently valuable, list a few related research papers.